

$\left. \begin{array}{l} \uparrow x \uparrow y \uparrow \\ \downarrow x \uparrow y \downarrow \end{array} \right\} \text{How?} \Rightarrow \text{Pearson's correlation coeff}^n$
 $\text{Cov}(X, Y)$

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

I know that

Ex: $\uparrow$

$x \uparrow \quad y \uparrow$

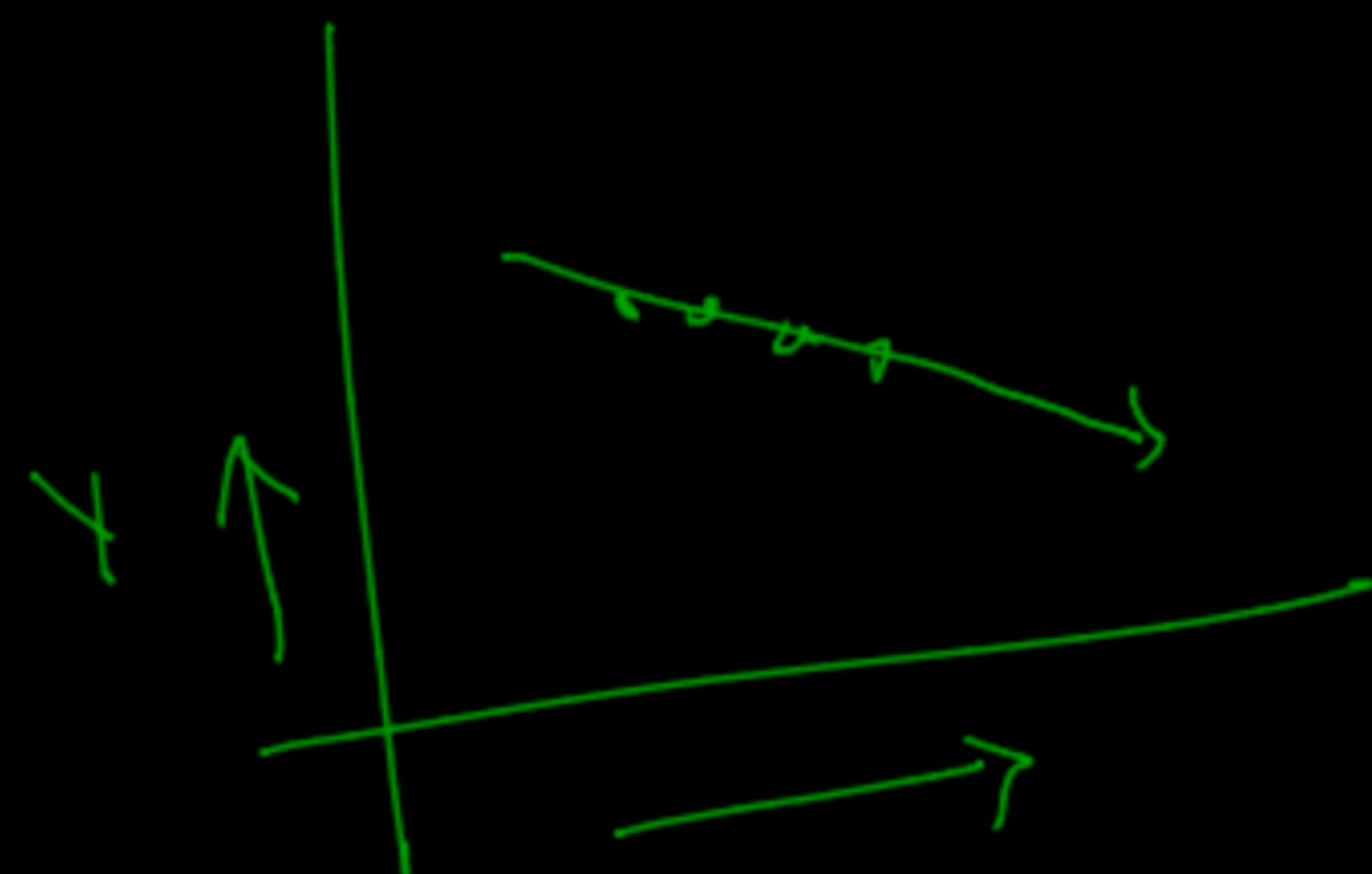
height, weight height weight

$\text{Cov}(X,Y) \rightarrow (+ve)$ But $\underline{\underline{+ \uparrow}}$

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_h \cdot \sigma_w}$$

① If $\boxed{\rho = -1} \rightarrow \text{strictly decreasing}$

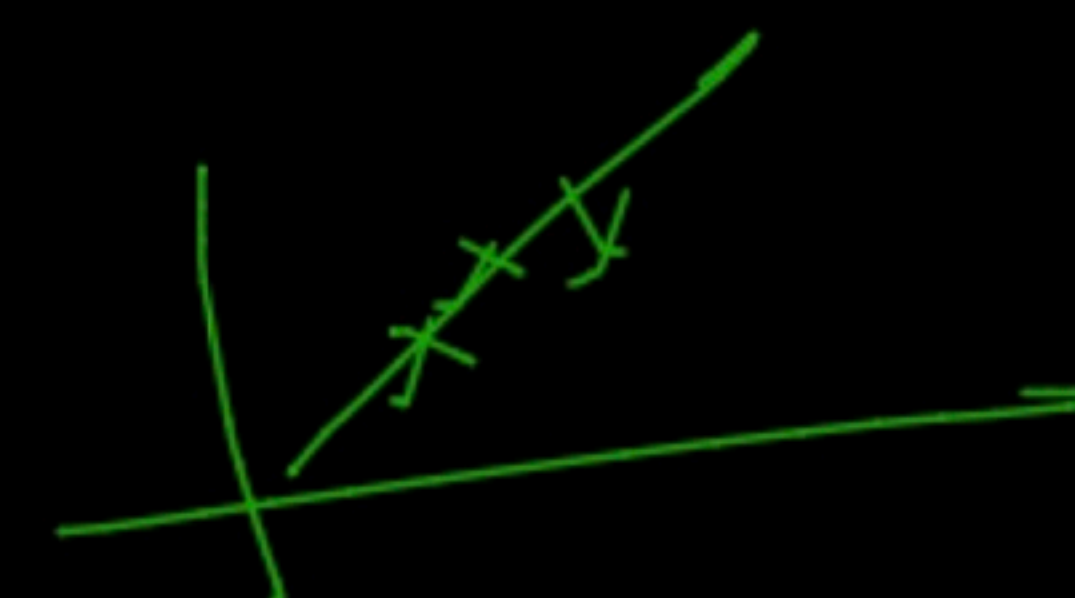
$$\rho = -1$$



(11)

$$0 < \rho < +1$$

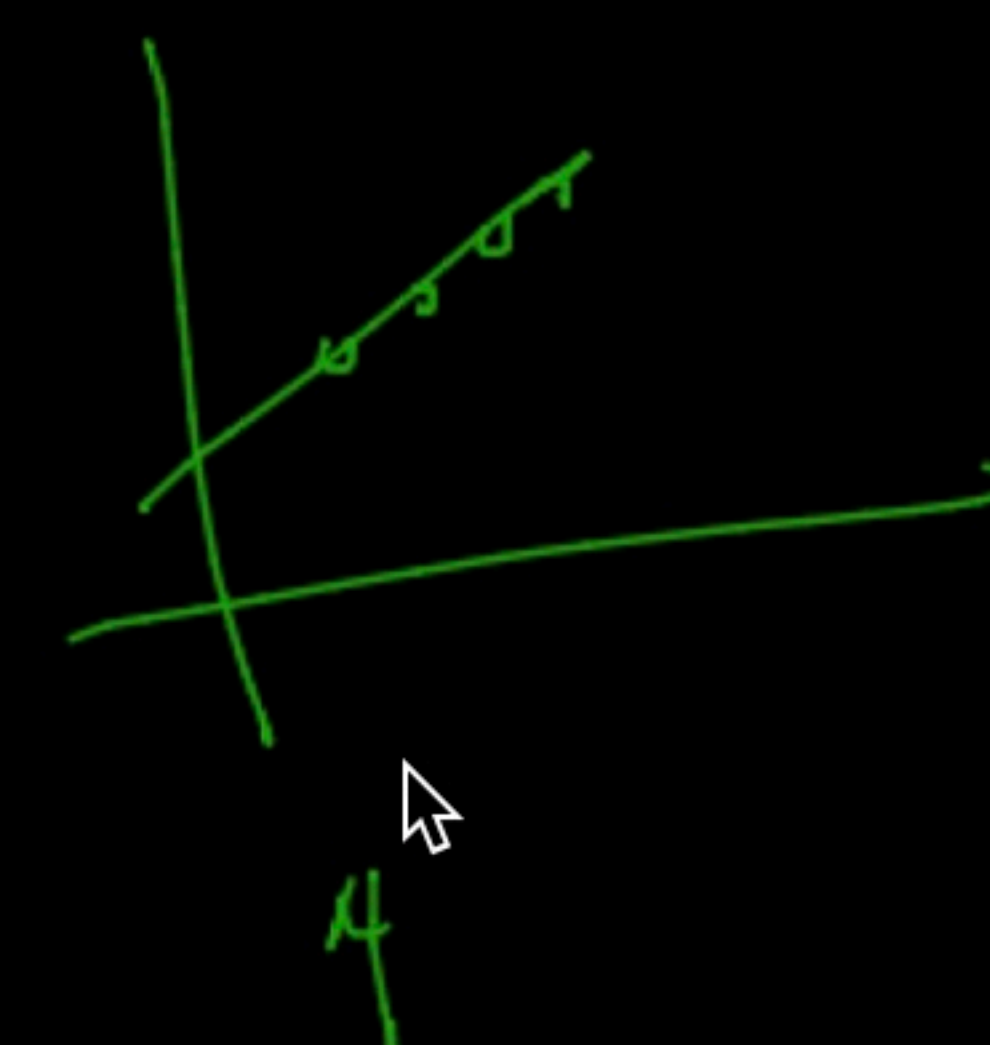
$\hookrightarrow$



(12)

$$\rho = +1$$

$\downarrow$

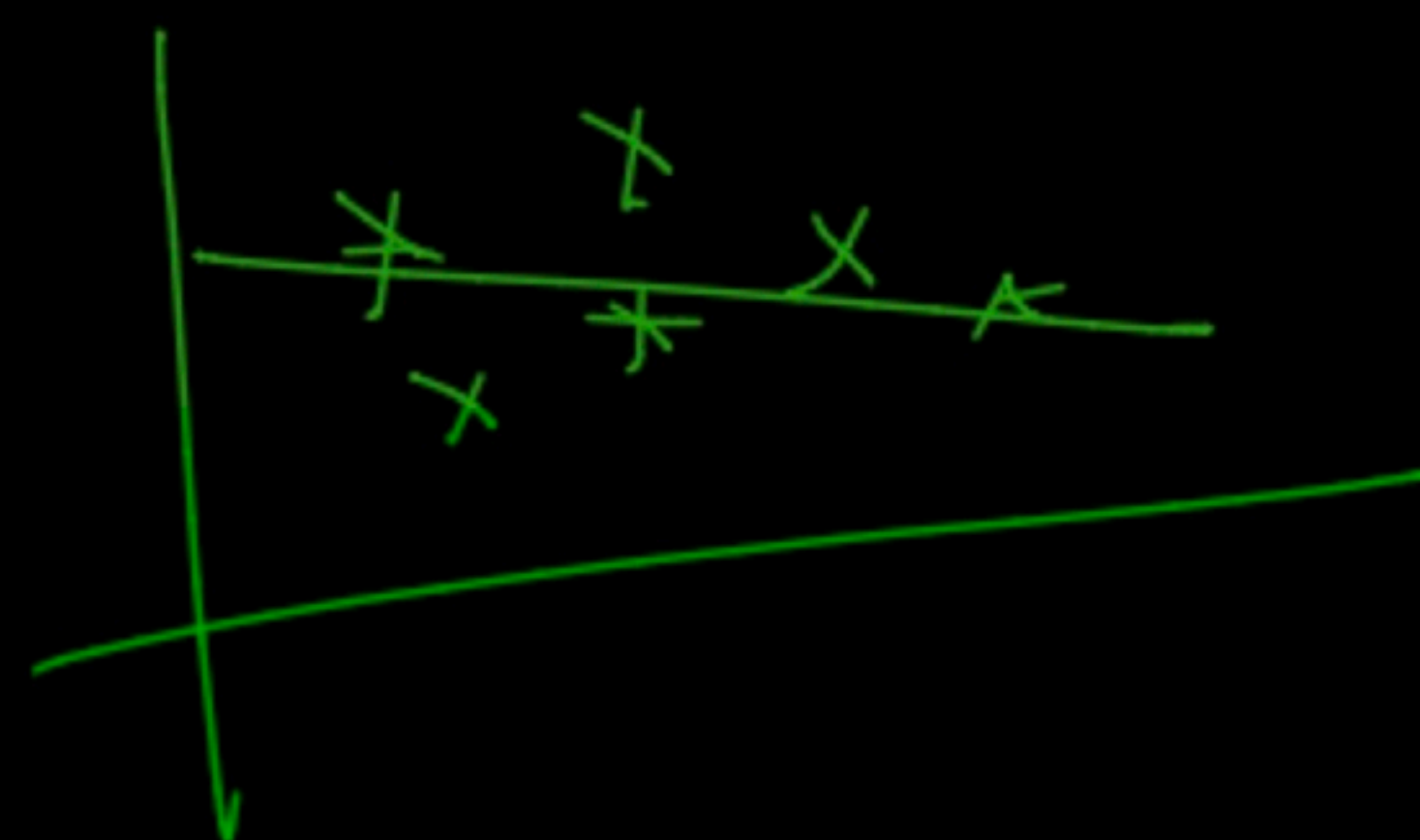


(13)

$$-1 < \rho < 0$$

$\hookrightarrow$

Not perfect



ii) Spearman's Rank co-relation

↓

Height	weight	x_h	x_w
160	52	4	3
150	66	2	4
170	68	5	5
140	46	1	1
158	51	3	2

$$r = 1$$

$$(x \uparrow \quad x \uparrow)$$

$$r = -1$$

$$(x \uparrow \quad x \downarrow)$$

$$r = \rho_{x_h, x_w}$$

Rank of
height

Rank of weight

$$\frac{cov(x_h, x_w)}{\sigma_{x_h} \sigma_{x_w}}$$

$X \uparrow Y \uparrow$

"correlation does not imply causation"

$\Downarrow$

$X \text{ cause } Y$
 $Y \text{ cause } X$

Hypothesis testing

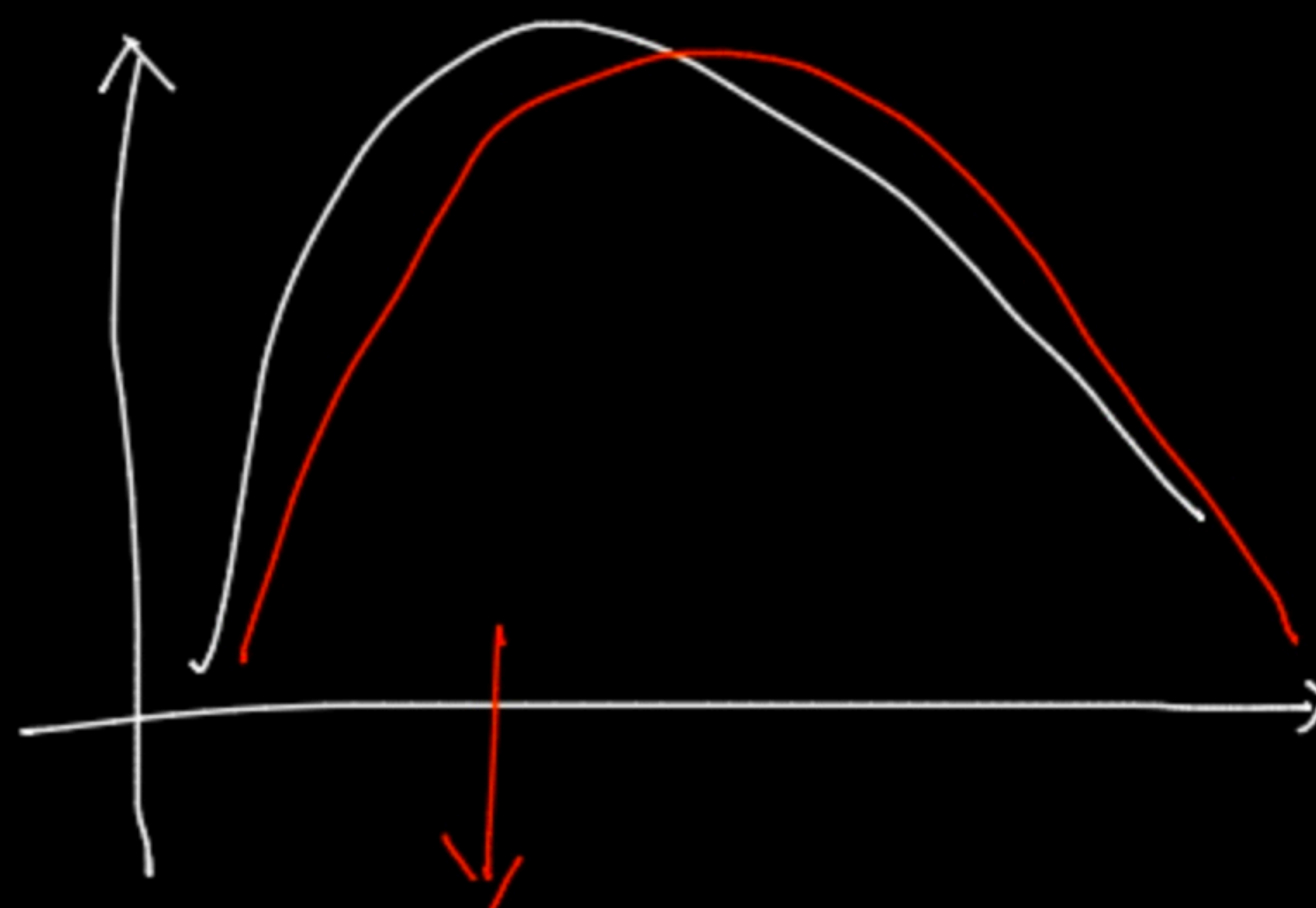
Q) Is there any diffⁿ b/w height of (C) and

	Class A	Class B
1 →	160	162
	162	156
50 →	148	182

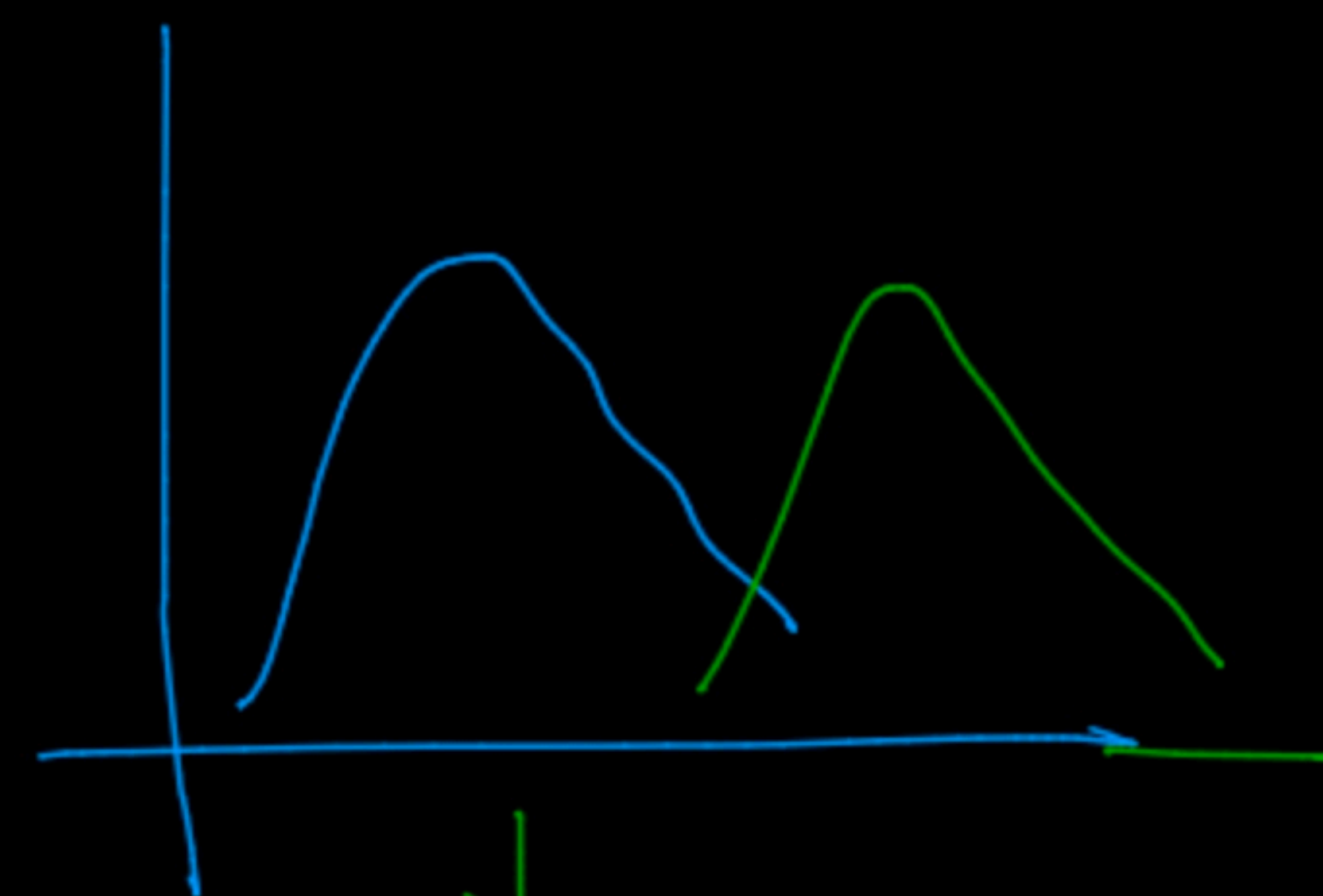
Class B
How will you solve it

↓

(case 1)



same



diffⁿ

↓

'Not same'

small → inference
conclude

Step 1: Test statistics

$$\mu_x - \mu_y = 0$$

Step 2: Define Null hypothesis $\rightarrow H_0$

$H_0 \rightarrow$ No diffⁿ in height

$H_1 \rightarrow$ inverse of H_0
ie. there is some diffⁿ

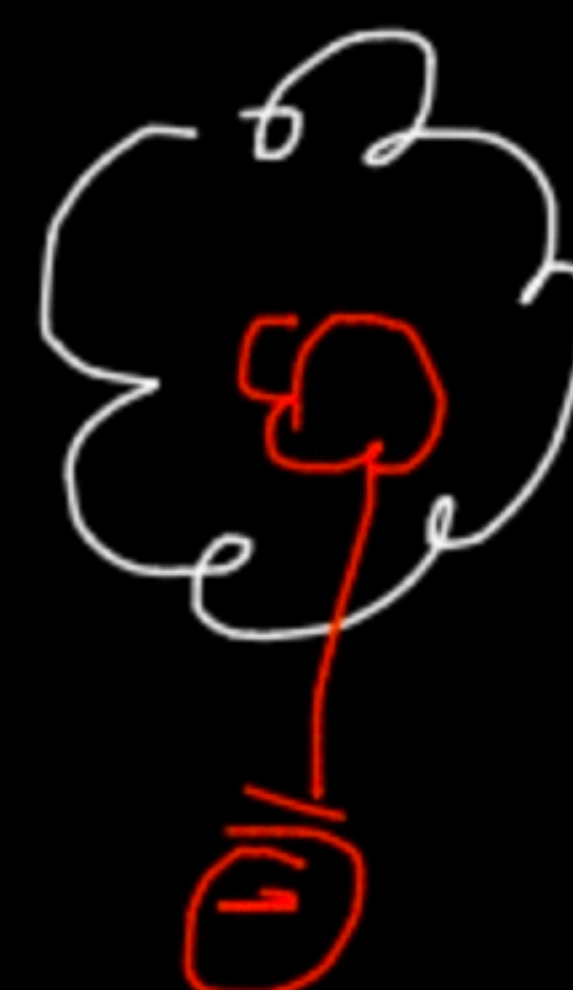
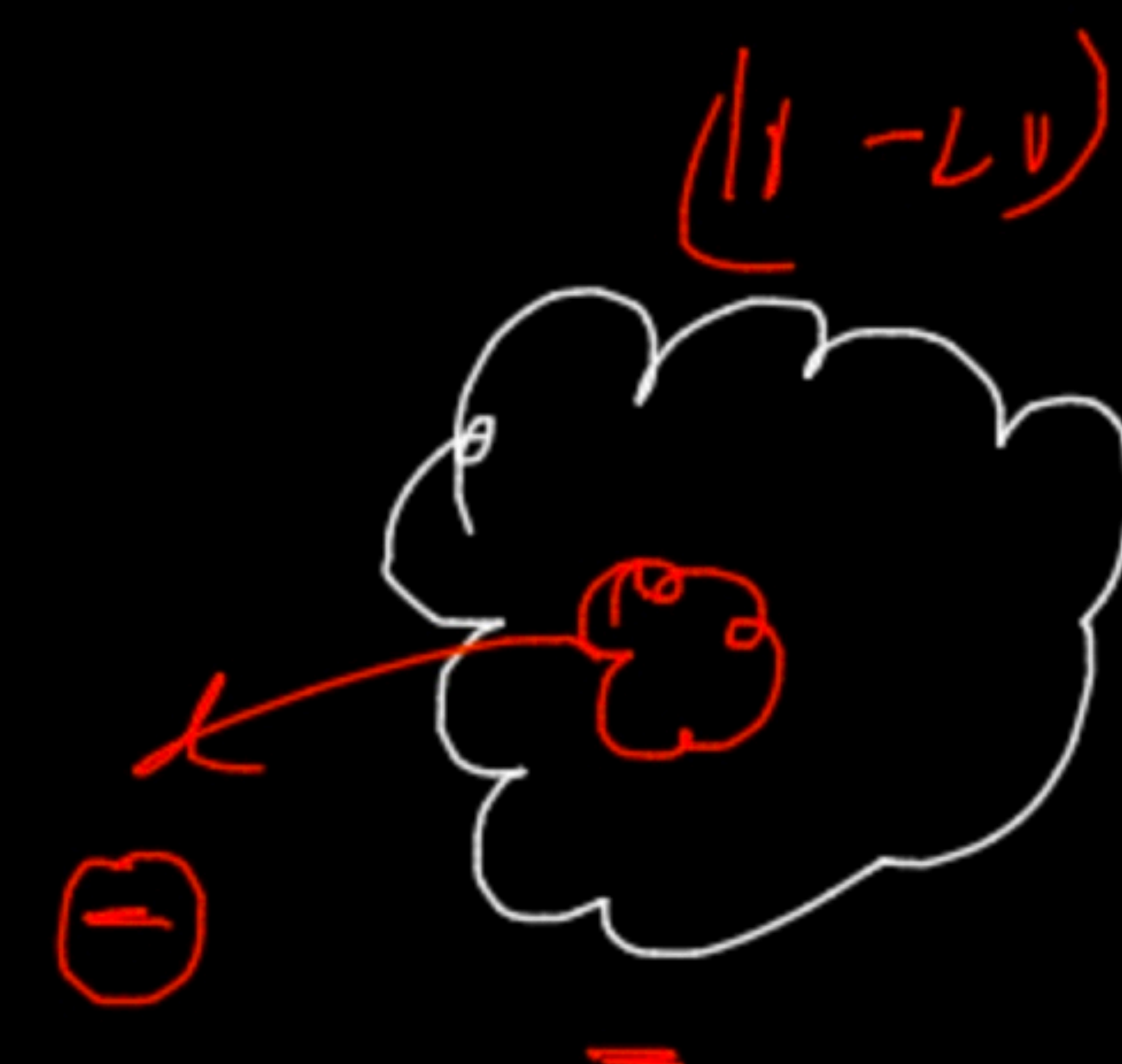
Step 3 compute P-value

what is the prob
of observing your
value
ie $(\mu_2 - \mu_1)$ if
my Null Hypothesis
is true

No diffⁿ

P-value

$$\frac{17}{20}$$



$\rightarrow 0$
 $\uparrow$

$\hookrightarrow v_{10} \rightarrow \text{Reject}$

